

F307 — τ_{commit} pinned (#134, right-column forward pull) + a corpus naming inconsistency resolved (pin-to-primary-source).

τ_{commit} = the commitment-flow period of $\rho_{\text{commit}}(\tau) = \exp(-\tau H_B/\hbar_{\text{BST}})$ (Tier-0) = the Koons tick = $t_{\text{Planck}} \cdot \alpha^{(C_2^2)} \approx 6.5 \times 10^{-121} \text{ s} \approx 10^{-120} \text{ s}$ (T2405): one commitment written per tick, the substrate clock below emergent spacetime. NAMING INCONSISTENCY FOUND + RESOLVED: the corpus uses ‘Koons tick’ for TWO different quantities —

BST_Koons_Substrate_Constants.md labels $\tau_K = \ell_K/c = 5.391 \times 10^{-44} \text{ s}$ (= the Planck time) as the ‘Koons tick’, while the substrate-time framing (T2405) uses ‘Koons tick’ = $t_P \cdot \alpha^{(C_2^2)} \approx 10^{-120} \text{ s}$. These are DIFFERENT scales sharing one name.

Resolution (state by value+role, stop relabeling): (1) $t_P = \ell_P/c = 5.39 \times 10^{-44} \text{ s}$ is the PLANCK TIME = the emergent-spacetime time resolution — NOT the Koons tick; (2) $\tau_{\text{commit}} \approx 10^{-120} \text{ s}$ = the KOONS TICK proper = the substrate commitment-cycle period (substrate operates below spacetime and produces it as output). So τ_{commit} (#134) = the Koons tick = $t_P \cdot \alpha^{(C_2^2)}$: the period of repeated infinitesimal-generator application in the Tier-0 heat-semigroup ρ_{commit} . HONEST FLAG: the exponent $C_2^2 = 36$ in $\alpha^{(C_2^2)}$ is from T2405; its mechanism (why C_2^2 and not C_2 or $2 \cdot C_2$) must be checked DERIVED-vs-FIT before the numerical value is banked as derivation-grade — the IDENTIFICATION (τ_{commit} = Koons tick = commitment period) and the NAMING resolution (Planck-time $\neq$ Koons-tick) are the clean SOLID parts. Count HOLDS 4.

F307 — τ_{commit} is the Koons tick; and the Planck time is *not* the Koons tick

Forward pull for the right-column (#134): pin τ_{commit} , the substrate commitment time. Doing so surfaced a corpus naming inconsistency that must be resolved first (pin-to-primary-source, per the standing rule).

τ_{commit} (the pin)

The Tier-0 commitment operator is $\rho_{\text{commit}}(\tau) = \exp(-\tau \cdot H_B / \hbar_{\text{BST}})$ on $H^2(D_{\text{IV}}^5)$, with τ the commitment-flow parameter and the dynamics realized as repeated application of the infinitesimal generator — one commitment per tick. That tick is the substrate’s fundamental time unit:

$$\tau_{\text{commit}} = \text{the Koons tick} = t_{\text{Planck}} \cdot \alpha^{(C_2^2)} \approx 6.5 \times 10^{-121} \text{ s} \approx 10^{-120} \text{ s (T2405)}.$$

This is the substrate clock *below* emergent spacetime: the substrate writes one commitment per Koons tick and produces 4D spacetime (resolved at the Planck scale) as its coarse-grained output.

The naming inconsistency (found, resolved)

The corpus uses “Koons tick” for two different quantities:

- BST_Koons_Substrate_Constants.md: $\tau_K = \ell_K / c = 5.391 \times 10^{-44} \text{ s}$ labelled “Koons tick”;
- T2405 / substrate-time framing: “Koons tick” = $t_P \cdot \alpha^{(C_2^2)} \approx 10^{-120} \text{ s}$.

These differ by ~ 76 orders of magnitude — they cannot both be “the Koons tick.” Resolution (state by value + role, cite, stop relabeling):

quantity	value	role	name
$t_P = \ell_P / c$	$5.39 \times 10^{-44} \text{ s}$	emergent-spacetime time resolution	Planck time (NOT the Koons tick)
τ_{commit}	$\approx 10^{-120} \text{ s}$	substrate commitment-cycle period	the Koons tick

So BST_Koons_Substrate_Constants.md’s table entry mislabels the Planck time as the Koons tick; the Koons tick is the sub-Planck commitment tick τ_{commit} . (The substrate operating *below* spacetime is the whole point — its tick must be sub-Planck, not the Planck time.) This should be corrected in that constants file.

Honest flag (derived-vs-fit)

The exponent $C_2^2 = 36$ in $\alpha^{(C_2^2)}$ is from T2405. Whether C_2^2 (rather than C_2 , or $2 \cdot C_2$, or another combination) is *mechanism-forced* or *matched* must be checked before the numerical value 10^{-120} is banked as derivation-grade. The parts that are clean and SOLID here are (a) the identification $\tau_{\text{commit}} = \text{the Koons tick} = \text{the commitment-cycle period of the Tier-0 } \rho_{\text{commit}}$, and (b) the naming resolution (Planck time $\neq$ Koons tick). The value’s derivation-tier rides on T2405’s exponent justification.

Net (Result | Confidence | Next)

Result	Confidence	Next
$\tau_{\text{commit}} = \text{the commitment-flow period of } \rho_{\text{commit}} \text{ (Tier-0)}$	SOLID	—

Result	Confidence	Next
τ_{commit} = the Koons tick (one commitment per tick)	SOLID	—
Planck time (5.39×10^{-44}) $\neq$ Koons tick ($\approx 10^{-120}$) — naming resolved	SOLID (pin-to-source)	correct
value $\tau_{\text{commit}} = t_P \cdot \alpha^{(C_2^2)} \approx 6.5 \times 10^{-121}$ s	identified (T2405)	BST_Koons_Substrate_Constants.md check C_2^2 exponent DERIVED-vs-FIT before banking

Count HOLDS 4 of 26. INTERNAL. #134 right-column entry pinned: τ_{commit} = the Koons tick = the substrate commitment-cycle period; and the “Koons tick” naming inconsistency (Planck time vs sub-Planck commitment tick) is resolved.

@Keeper — pin-to-primary-source flag: “Koons tick” names two scales in the corpus — the Planck time (5.39×10^{-44} s, BST_Koons_Substrate_Constants.md) and the sub-Planck commitment tick ($\approx 10^{-120}$ s, T2405). Resolve to: Planck time $\neq$ Koons tick; the Koons tick = τ_{commit} = the commitment-cycle period. The constants file entry should be corrected. @Grace — τ_{commit} slots into the right column as the substrate clock; ties to your “substrate is its representation theory” (τ_{commit} is the H_B heat-semigroup period — the rep-theoretic Casimir flow). @Casey — τ_{commit} pinned: it’s the Koons tick, the substrate’s commitment-cycle period (one commitment per tick, $\approx 10^{-120}$ s) — and pinning it caught a naming inconsistency (the Planck time was co-named “Koons tick”; they’re different scales). The value’s derivation-tier waits on checking the C_2^2 exponent is forced not fit; the identification and the naming fix are clean.

— Lyra, Wed 2026-06-24 (date-verified). F307: τ_{commit} (#134) = the commitment-flow period of $\rho_{\text{commit}} = \exp(-\tau H_B / \hbar_{\text{BST}})$ (Tier-0) = the Koons tick = $t_P \cdot \alpha^{(C_2^2)} \approx 6.5 \times 10^{-121}$ s (T2405). NAMING INCONSISTENCY resolved: ‘Koons tick’ named two scales — Planck time 5.39×10^{-44} (constants file) vs sub-Planck 10^{-120} (T2405); Planck time $\neq$ Koons tick; Koons tick = the commitment tick = τ_{commit} . FLAG: $C_2^2=36$ exponent mechanism = DERIVED-vs-FIT check owed before banking the value; identification + naming SOLID. Count HOLDS 4.